Multiple Choice Questions: L.C.M and H.C.F

Mathematics Practice Module

December 30, 2025

1 Practice Questions

Q1. The H.C.F. of two numbers is 16 and their L.C.M. is 160. If one of the numbers is 32, then the other number is:
A) 48 B) 80 C) 96 D) 112

Q2. Which of the following is a pair of co-primes?
A) (16, 62) B) (18, 25) C) (21, 35) D) (23, 92)

Q3. The L.C.M. of $\frac{1}{3}, \frac{5}{6}, \frac{2}{9}, \frac{4}{27}$ is:
A) $\frac{1}{54}$ B) $\frac{10}{27}$ C) $\frac{20}{3}$ D) None of these

Q4. The H.C.F. of $2^2 \times 3^3 \times 5^5$, $2^3 \times 3^2 \times 5^2 \times 7$ and $2^4 \times 3^4 \times 5 \times 7^2 \times 11$ is:
A) $2^2 \times 3^2 \times 5$ B) $2^2 \times 3^2 \times 5 \times 7 \times 11$ C) $2^4 \times 3^4 \times 5^5$ D) $2^4 \times 3^4 \times 5^5 \times 7^2 \times 11$

Q5. The ratio of two numbers is 3 : 4 and their H.C.F. is 4. Their L.C.M. is:
A) 12 B) 16 C) 24 D) 48

Q6. The smallest number which when diminished by 7, is divisible by 12, 16, 18, 21 and 28 is:
A) 1008 B) 1015 C) 1022 D) 1032

Q7. The H.C.F. of two numbers is 11 and their L.C.M. is 7700. If one of the numbers is 275, then the other is:
A) 279 B) 283 C) 308 D) 318

Q8. The greatest number that will divide 187, 233 and 279 so as to leave the same remainder in each case is:
A) 30 B) 36 C) 46 D) 56

Q9. The L.C.M. of two numbers is 48. The numbers are in the ratio 2 : 3. The sum of the numbers is:
A) 28 B) 32 C) 40 D) 64

Q10. Find the H.C.F. of $\frac{9}{10}, \frac{12}{25}, \frac{18}{35}$ and $\frac{21}{40}$.
A) $\frac{3}{5}$ B) $\frac{252}{5}$ C) $\frac{3}{1400}$ D) $\frac{63}{700}$

Q11. Six bells commence tolling together and toll at intervals of 2, 4, 6, 8, 10 and 12 seconds respectively. In 30 minutes, how many times do they toll together?
A) 4 B) 10 C) 15 D) 16

Q12. The H.C.F. of two numbers is 12 and their difference is 12. The numbers are:
A) 66, 78 B) 70, 82 C) 84, 96 D) 94, 106

Q13. The product of two numbers is 2028 and their H.C.F. is 13. The number of such pairs is:
A) 1 B) 2 C) 3 D) 4

Q14. The least number which when divided by 5, 6, 7 and 8 leaves a remainder 3, but when divided by 9 leaves no remainder, is:
A) 1683 B) 1692 C) 2523 D) 3363

Q15. The H.C.F. and L.C.M. of two numbers are 8 and 48 respectively. If one of the numbers is 24, then the other number is:
A) 48 B) 36 C) 24 D) 16

Q16. The greatest number of four digits which is divisible by 15, 25, 40 and 75 is:
A) 9000 B) 9400 C) 9600 D) 9800

Q17. The L.C.M. of 22, 54, 108, 135 and 198 is:
A) 330 B) 1980 C) 5940 D) 11880

Q18. Three numbers are in the ratio 1 : 2 : 3 and their H.C.F. is 12. The numbers are:
A) 4, 8, 12 B) 5, 10, 15 C) 10, 20, 30 D) 12, 24, 36

Q19. The least number which is a perfect square and is divisible by each of the numbers 16, 20 and 24 is:
A) 1600 B) 3600 C) 6400 D) 14400

Q20. The H.C.F. of two numbers is 8. Which one of the following can never be their L.C.M.?
A) 24 B) 48 C) 56 D) 60

Q21. The L.C.M. of two numbers is 495 and their H.C.F. is 5. If the sum of the numbers is 100, then their difference is:
A) 10 B) 46 C) 70 D) 90

Q22. Find the greatest number which on dividing 1657 and 2037 leaves remainders 6 and 5 respectively.
A) 123 B) 127 C) 235 D) 305

Q23. The L.C.M. of two numbers is 20 times their H.C.F. The sum of H.C.F. and L.C.M. is 2520. If one of the numbers is 480, the other number is:
A) 400 B) 480 C) 520 D) 600

Q24. The capacity of two pots are 120 litres and 56 litres respectively. Find the capacity of a container which can exactly measure the contents of the two pots.
A) 7500 cc B) 7850 cc C) 8000 cc D) 9500 cc

Q25. The product of two numbers is 4107. If the H.C.F. of these numbers is 37, then the greater number is:
A) 101 B) 107 C) 111 D) 185

Q26. Find the lowest common multiple of 24, 36 and 40.
A) 120 B) 240 C) 360 D) 480

Q27. The H.C.F. of two numbers is 12 and their L.C.M. is 72. If the sum of the numbers is 60, then one of the numbers is:
A) 12 B) 24 C) 60 D) 72

Q28. Find the greatest number of five digits which is exactly divisible by 12, 15, 18 and 27.
A) 99600 B) 99720 C) 99900 D) 99350

Q29. The H.C.F. of 0.54, 1.8 and 7.2 is:
A) 0.18 B) 1.8 C) 0.018 D) 18

Q30. A rectangular courtyard 3.78 m long and 5.25 m wide is to be paved exactly with square tiles, all of the same size. What is the largest size of the tile which could be used?
A) 14 cm B) 21 cm C) 42 cm D) 63 cm

2 Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	7	C	13	B	19	B	25	C
2	B	8	C	14	A	20	D	26	C
3	C	9	C	15	D	21	A	27	B
4	A	10	C	16	C	22	B	28	B
5	D	11	D	17	C	23	D	29	A
6	B	12	C	18	D	24	C	30	B

3 Brief Explanatory Hints

- **Q11:** LCM of (2,4,6,8,10,12) is 120s = 2 mins. In 30 mins, they toll $30/2 + 1 = 16$ times (including the start).
- **Q13:** Let numbers be $13a$ and $13b$. $13a \times 13b = 2028 \Rightarrow ab = 12$. Co-prime pairs for (a, b) are (1,12) and (3,4).
- **Q21:** $x + y = 100$, $xy = 495 \times 5 = 2475$. $(x - y)^2 = (x + y)^2 - 4xy$.
- **Q24:** HCF of 120 and 56 is 8 Litres. 1 Litre = 1000 cc. So, 8000 cc.
- **Q30:** Convert to cm (378 and 525). Find HCF of 378 and 525.